Supplementary material: conditioning the aging hazard and validating the chimera-death object

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This supplement reports two run-first robustness analyses prompted by the two critical questions a reader may raise about the headline statistical claims: (C1) whether the $A = 0.5$ aging is a genuine state-level hazard or an artifact of pooling fixed transient times across the seed ensemble, and (C2) whether the nonlocal-ring chimera-death study's increasing-hazard result is contaminated by a non-canonical decay channel or by the choice of death criterion. Every number below is regenerated by committed scripts (`critical_review/` in the repository) and was independently reproduced with a second, disjoint censored-Weibull implementation; no figure or value is a placeholder. Section 1 concerns the two-population results of the main text directly; Section 2 concerns the nonlocally-coupled ring (Wolfrum–Omel'chenko) chimera-death campaign. Section 3 lists the short main-text sentences these analyses license.

1 Conditioning the $A = 0.5$ aging on the collective initial condition

At the operating corner the reduced flow is deterministic, so a censored-Weibull shape $k_{\mathrm{abs}} > 1$ could in principle encode only the *spread of fixed transient times* across the random-seed ensemble (an initial-condition heterogeneity), rather than a hazard that rises along each trajectory. We separate the two by stratifying the 200-seed ensemble at each population size into four equal-count bins of a collective initial-condition coordinate and refitting the right-censored Weibull *within* each stratum. We use three independent stratifying variables: the initial inter-population phase gap $|\Delta\phi_0|$; the initial incoherent-population order parameter $\overline{R}_{\mathrm{incoh},0}$; and, as the decisive test, the reduced three-variable model's *predicted* lifetime t_{cap} for that seed—which collapses the entire collective initial condition onto its lifetime consequence, so that conditioning on it removes every lifetime dependence the collective coordinates can express.

The aging survives conditioning in every case (Table 1, Fig. 1). The pooled per-N fits reproduce the main-text values $k_{\mathrm{abs}} = 2.27, 2.22, 2.49, 2.67, 2.44, 2.72, 3.03$ exactly. Within strata, $k_{\mathrm{abs}} > 1$ with 95% profile-likelihood confidence intervals excluding one in all 28 (size $\times$ bin) cells for each of the three variables—$84/84$ cells in total—with k_{abs} ranging from 1.88 to 5.12. The within-bin coefficient of variation of τ_{abs} remains substantial (0.22–0.56, not a degenerate spike) and tracks the fitted shape through the Weibull relation $\mathrm{CV}(k) = \sqrt{\Gamma(1+2/k)/\Gamma(1+1/k)^2 - 1}$ (empirical-vs-predicted correlation 0.996), confirming a genuine aging-shaped within-stratum lifetime spread. Conditioning on the reduced model's predicted lifetime—the strongest control available—leaves $k_{\mathrm{abs}} \in [2.00, 5.00]$, again excluding one in all 28 cells; within narrow predicted-lifetime bands the shape if anything *grows* with N, the opposite of a collective-IC confound. This is consistent with the per-cycle ratchet of Sec. 5 (the per-pass hazard increment rises from 0.031 to 0.091 across N, with the fraction of monotone-ratcheting runs reaching unity by $N = 32$): the hazard rises as the chimera ages regardless of where it started. We note that at $A = 0.5$ the censored fraction is exactly zero (all 1400 runs absorb before $t_{\max} = 2000\,\mathrm{s}$), so these are effectively uncensored maximum-likelihood fits.

Table 1: Within-stratum aging at $A = 0.5$. For each stratifying variable the seed ensemble is split into four equal-count bins per N; a right-censored Weibull is fit within each (size, bin) cell. "CI excl. 1" counts cells whose 95% profile-likelihood interval on k_{abs} lies above the memoryless value $k = 1$.

stratifying variable	cells	k_{abs} range	CI excl. 1
(pooled per-N, reference)	7	2.22–3.03	7/7
$\lvert\Delta\phi_0\rvert$ (phase gap)	28	1.88–5.12	28/28
t_{cap} (reduced pred. lifetime)	28	2.00–5.00	28/28
$\overline{R}_{\mathrm{incoh},0}$ (incoherent IC level)	28	1.88–4.83	28/28

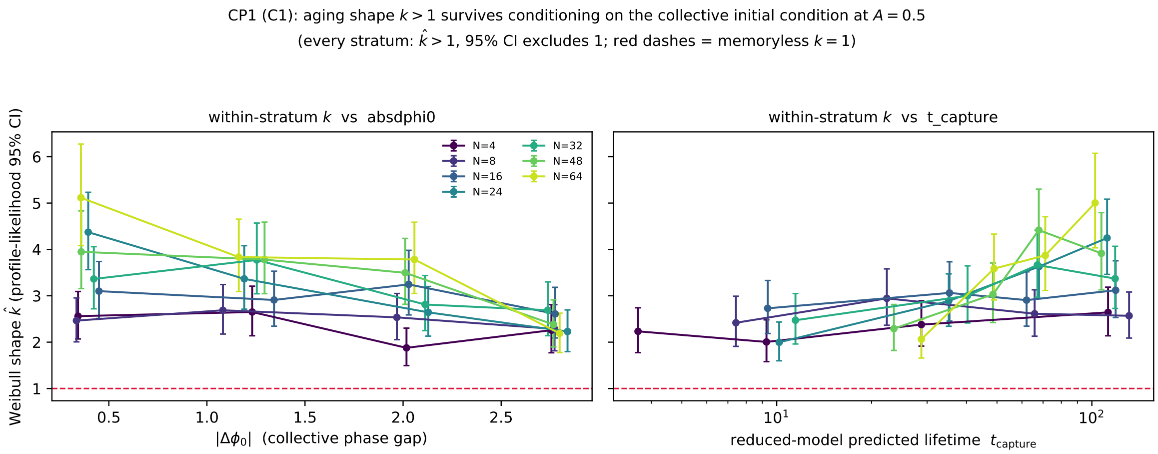

Figure 1: Aging survives conditioning on the collective initial condition at $A = 0.5$. Within-stratum censored-Weibull shape $\hat{k}$ (profile-likelihood 95% intervals) versus bin, one curve per population size N, for stratification by $|\Delta\phi_0|$ (left) and by the reduced model's predicted lifetime t_{cap} (right). Every stratum sits above the memoryless value $k = 1$ (red dashes); within a fixed collective initial condition the lifetime distribution keeps an increasing-hazard shape, so the aging is a property of the trajectory, not of the initial-condition ensemble.

2 The dying ring object and its criterion-independence

The nonlocal-ring chimera-death campaign reports a Weibull shape that saturates to a finite $k_\infty \approx 1.35$–$1.65 > 1$ as $N \to \infty$ (per-β bootstrap intervals excluding one). Two features invite scrutiny. First, the median ring lifetime *decreases* with N in the studied window—opposite to the lifetime growth of the deep-stable regime [Wolfrum & Omel'chenko, Phys. Rev. E **84**, 015201(R) (2011)]—raising the worry that the death criterion ($\rho_{\mathrm{std}}(t) < 0.04$ held for 50 time units, where ρ_{std} is the spatial standard deviation of the local order parameter ρ_k) times a non-canonical decay channel. Second, the increasing-hazard shape might be specific to that criterion. We address both directly; both survive.

The object being killed is a canonical chimera (97–98%). The spatial field $\rho_k(t,\cdot)$ was not stored, but every run is exactly reproducible from its logged seed, so we re-ran all 300 trajectories per cell at $N = 256$ for $\beta = 0.110$ and $\beta = 0.130$ with field dumping; the re-detected lifetimes reproduce the campaign values bit-for-bit (0/600 mismatches). Classifying each pre-death state by its spatial structure (a contrast gate on ρ_{std} together with the dominant spatial Fourier mode of ρ_k, which is drift invariant) gives the breakdown in Table 2. The death criterion times the collapse of a genuine chimera—one contiguous coherent arc plus

Table 2: Pre-death object at $N = 256$ (300 runs per cell). "chimera death" is the sum of single-arc and multi-headed chimera collapses. $\hat{k}$ columns are right-censored Weibull shapes (profile-likelihood 95% intervals) on all runs, on chimera deaths only, and on single-arc deaths only.

	single-arc	multi-head	degen.	chim. death	$\hat{k}$ all	$\hat{k}$ chim.	$\hat{k}$ single
$\beta = 0.110$	83.7%	14.3%	2.0%	98.0%	1.22 [1.12, 1.32]	1.24 [1.14, 1.35]	1.24 [1.13, 1.36]
$\beta = 0.130$	89.3%	8.0%	2.7%	97.3%	1.47 [1.35, 1.58]	1.48 [1.36, 1.60]	1.46 [1.34, 1.59]

Table 3: Criterion-independence at $\beta = 0.130$. Median lifetime $\tilde{\tau}$ (left) and Weibull shape $\hat{k}$ (right) versus N, re-detected on the same trajectories under each criterion. "mean-coh (corr.)" re-times the twisted-state collapses the raw mean-coherence rule mislabels as censored. All ρ_{std}-based criteria carry 0% censoring.

	median $\tilde{\tau}(N)$					$\hat{k}(N)$				
criterion	32	64	128	192	256	32	64	128	192	256
$\rho_{\mathrm{std}} < 0.04$ (orig.)	872	422	404	349	345	1.33	1.55	1.68	1.68	1.47
$\rho_{\mathrm{std}} < 0.08$	828	383	372	314	308	1.28	1.45	1.61	1.56	1.42
$\rho_{\mathrm{std}} < 0.10$	815	372	356	303	289	1.26	1.41	1.59	1.50	1.40
mean-coh (corr.)	810	379	362	310	276	1.26	1.44	1.62	1.54	1.57

one incoherent arc, or a multi-headed variant—in 98.0% ($\beta = 0.110$) and 97.3% ($\beta = 0.130$) of runs; the degenerate "dissolve-then-linger" channel (the chimera disperses early and a low-contrast state persists until full homogenisation) is real but rare, 2.0–2.7%. Refitting after removing it leaves the Weibull shape unchanged—if anything slightly larger—with intervals still excluding one (Table 2); sweeping the contrast gate over $[0.06, 0.12]$ only moves $\hat{k}$ further from one. The k_∞ claim is therefore not an artifact of the rare degenerate channel.

The trend and the hazard shape are criterion-independent. Re-detecting death on the same $\beta = 0.130$ trajectories under alternative literature-style criteria, the median lifetime *decreases* with N under every one (Table 3, left; $\tilde{\tau}(256)/\tilde{\tau}(32) = 0.34$–$0.40$), so the inversion is a property of the near-boundary dynamics, not of the detector. The increasing-hazard shape $\hat{k}(N) > 1$ is likewise robust under the structure-loss criteria (ρ_{std} below 0.08 or 0.10; Table 3, right; all intervals exclude one, with 0% censoring). A mean-coherence-level criterion (ρ-mean above a fixed value) appears to drive $\hat{k}$ below one at large N, but this is a diagnosed censoring artifact: 6.7% of the $N = 256$ collapses go to a spatially uniform *twisted* state ($\rho_k \approx 0.50$, genuinely dead by $\rho_{\mathrm{std}} < 0.04$ at finite time) rather than the high-coherence sync plateau, and a mean-coherence threshold mislabels these genuine deaths as censored, injecting a spurious immortal tail; re-timing them at their true death restores $\hat{k}(N) = 1.26, 1.44, 1.62, 1.54, 1.57$, all intervals above one. An independent Kaplan–Meier log-cumulative-hazard slope (1.5–2.1 across N) corroborates the increasing hazard without any Weibull fit. That the spatial-homogeneity criterion—unlike a coherence-level threshold—captures both the synchronised and the twisted collapse channels is itself a reason to prefer it.

3 Suggested main-text insertions

These analyses license the following short additions, kept deliberately minimal.

For Sec. 4 (Aging), after the k_{abs} values. "This aging is not an artifact of pooling heterogeneous deterministic transients: stratifying the seed ensemble by the collective initial condition—the inter-population

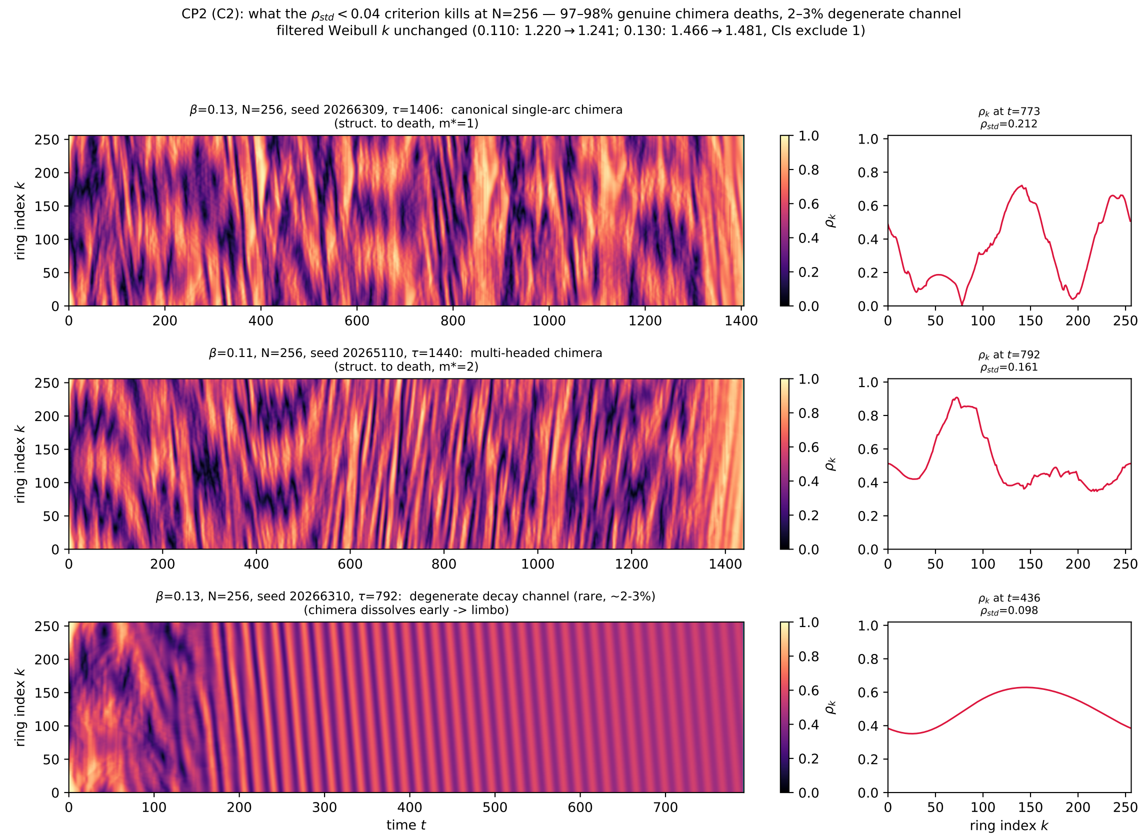

Figure 2: What the $\rho_{\mathrm{std}} < 0.04$ criterion kills at $N = 256$. Space–time local-coherence fields $\rho_k(t,k)$ (left) with a representative mature snapshot (right) for the three pre-death behaviours: a canonical single-arc chimera (top, structured to death), a multi-headed chimera (middle), and the rare degenerate channel (bottom, the chimera disperses early and a low-contrast state lingers). 97–98% of deaths are of the first two (genuine chimera) type; removing the degenerate channel leaves the Weibull shape unchanged.

phase gap, the incoherent-population level, or the reduced model's predicted lifetime—and refitting within each stratum leaves $k_{\mathrm{abs}} > 1$ with confidence intervals excluding one in all 84 size×stratum cells, and the within-stratum lifetime spread retains an increasing-hazard shape (Supplement, Sec. 1)."

For the ring-reframe / Discussion treatment of the nonlocal ring. "The decreasing ring median lifetime with N at these near-boundary β reflects the near-collapse side of the existence boundary rather than the deep-stable regime of Wolfrum–Omel'chenko, in which lifetimes grow with N; the dying object is a canonical (single- or multi-arc) chimera in 97–98% of runs, and the increasing-hazard shape $k(N) > 1$ is robust to the death criterion and to the rare degenerate decay channel (Supplement, Sec. 2)."

Methods/Data-availability note. "Stratified-hazard and criterion-robustness analyses, including the field re-runs validating the dying object, are in `critical_review/` and were independently reproduced with a disjoint censored-Weibull implementation."

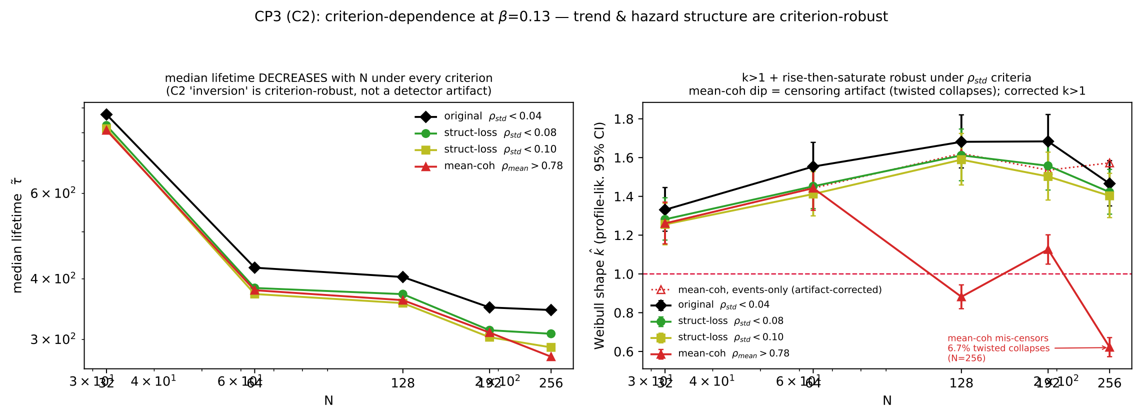

Figure 3: Criterion-independence at $\beta = 0.130$. Left: median lifetime decreases with N under every death criterion (the near-boundary inversion is not a detector artifact). Right: the Weibull shape $\hat{k}(N)$ stays above one under the spatial-homogeneity ($\rho_{\rm std}$) criteria; the mean-coherence rule's apparent dip below one at large N is a censoring artifact (twisted-state collapses mislabelled as censored), and the artifact-corrected curve (open markers) returns above one.